

## CANBY, MN, USA

WMO: 722252

Lat: 44.729N

Lon: 96.266W

Elev: 1193

StdP: 14.07

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 54923

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -10.5      | -7.0 | -17.3                         | 2.2 | -9.8 | -13.7 | 2.7 | -6.6 | 29.7                  | 23.0 | 26.9 | 17.9 | 6.3                   | 290  | 0.575 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.1                   | 91.0            | 72.8 | 88.4 | 71.8 | 85.8 | 70.6 | 77.3                | 87.0 | 75.2 | 84.4 | 73.2 | 81.9 | 10.4                 | 200  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 74.3                            | 133.9 | 83.4 | 72.3 | 125.0 | 81.6 | 70.3 | 116.7 | 79.2 | 41.6          | 86.9 | 39.5 | 83.9 | 37.6 | 81.8 | 83.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |       |                    | n-Year Return Period Values of Extreme Temperature |           |       |            |       |            |       |            |       |      |
|-------------------|------|------|----------------------------|-------|-------|--------------------|----------------------------------------------------|-----------|-------|------------|-------|------------|-------|------------|-------|------|
| 1%                |      | 2.5% | 5%                         | Mean  |       | Standard Deviation |                                                    | n=5 years |       | n=10 years |       | n=20 years |       | n=50 years |       |      |
| Min               |      | Max  |                            | Min   | Max   |                    | Min                                                | Max       | Min   | Max        | Min   | Max        | Min   | Max        |       |      |
| (a)               | (b)  | (c)  |                            | (e)   | (f)   | (g)                | (h)                                                | (i)       | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |
| 26.3              | 23.6 | 20.3 |                            | -18.5 | 96.9  | 5.1                | 3.6                                                | -22.2     | 99.5  | -25.2      | 101.6 | -28.1      | 103.6 | -31.8      | 106.3 |      |
|                   |      |      |                            | DB    |       |                    |                                                    |           |       |            |       |            |       |            |       |      |
|                   |      |      |                            | WB    | -18.9 | 79.8               | 5.0                                                | 1.8       | -22.6 | 81.1       | -25.5 | 82.1       | -28.3 | 83.2       | -32.0 | 84.5 |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 46.5   | 16.7  | 18.9  | 33.3  | 44.8  | 58.8  | 68.8  | 73.5  | 70.6  | 63.7  | 49.6  | 35.7  | 21.4  |
|                                                                                  | DBStd       | 22.58  | 14.10 | 13.00 | 13.85 | 10.48 | 9.02  | 6.42  | 5.57  | 5.60  | 8.16  | 9.01  | 11.81 | 12.51 |
|                                                                                  | HDD50       | 4138   | 1034  | 870   | 540   | 215   | 24    | 0     | 0     | 0     | 3     | 118   | 447   | 887   |
|                                                                                  | HDD65       | 7502   | 1499  | 1290  | 984   | 609   | 234   | 34    | 6     | 13    | 121   | 482   | 878   | 1352  |
|                                                                                  | CDD50       | 2848   | 0     | 0     | 22    | 58    | 298   | 564   | 728   | 638   | 415   | 106   | 19    | 0     |
|                                                                                  | CDD65       | 736    | 0     | 0     | 1     | 2     | 43    | 147   | 270   | 186   | 82    | 5     | 0     | 0     |
|                                                                                  | CDH74       | 7169   | 0     | 0     | 10    | 50    | 508   | 1398  | 2699  | 1622  | 807   | 73    | 2     | 0     |
|                                                                                  | CDH80       | 2416   | 0     | 0     | 1     | 10    | 160   | 467   | 987   | 520   | 254   | 17    | 0     | 0     |
| Wind                                                                             | WSAvg       | 8.8    | 9.7   | 10.0  | 9.6   | 10.4  | 9.3   | 7.8   | 6.5   | 6.2   | 7.9   | 9.4   | 9.7   | 9.3   |
| Precipitation                                                                    | PrecAvg     | 27.0   | 0.7   | 0.7   | 1.5   | 2.7   | 3.3   | 4.7   | 3.3   | 3.0   | 3.0   | 2.4   | 1.3   | 0.7   |
|                                                                                  | PrecMax     | 36.5   | 2.6   | 1.6   | 3.7   | 7.2   | 5.7   | 9.7   | 5.7   | 5.6   | 9.5   | 7.6   | 3.5   | 2.2   |
|                                                                                  | PrecMin     | 21.4   | 0.0   | 0.1   | 0.4   | 0.7   | 1.0   | 1.2   | 1.0   | 1.2   | 0.6   | 0.3   | 0.0   | 0.1   |
|                                                                                  | PrecStd     | 4.0    | 0.6   | 0.4   | 0.8   | 1.5   | 1.5   | 2.2   | 1.5   | 1.2   | 2.1   | 1.9   | 1.1   | 0.6   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 48.4  | 57.7  | 74.4  | 79.9  | 90.6  | 93.8  | 95.2  | 92.6  | 90.8  | 82.4  | 70.6  | 52.9  |
|                                                                                  |             | MCWB   | 39.7  | 50.1  | 57.6  | 58.0  | 65.5  | 75.0  | 74.5  | 74.6  | 70.0  | 62.9  | 55.6  | 45.2  |
|                                                                                  | 2%          | DB     | 42.9  | 46.8  | 65.4  | 73.4  | 84.1  | 89.5  | 91.2  | 89.1  | 86.9  | 74.9  | 63.8  | 46.5  |
|                                                                                  |             | MCWB   | 37.7  | 40.2  | 54.2  | 55.9  | 63.6  | 71.1  | 75.4  | 74.1  | 68.7  | 59.1  | 51.0  | 39.7  |
|                                                                                  | 5%          | DB     | 39.1  | 41.3  | 59.0  | 67.6  | 80.0  | 85.5  | 88.2  | 85.9  | 82.5  | 69.7  | 57.5  | 41.3  |
|                                                                                  |             | MCWB   | 35.4  | 37.1  | 50.5  | 51.6  | 62.4  | 70.3  | 72.8  | 71.9  | 67.3  | 56.7  | 47.9  | 36.3  |
|                                                                                  | 10%         | DB     | 36.0  | 37.1  | 53.0  | 62.5  | 74.8  | 81.8  | 85.7  | 82.3  | 78.7  | 64.4  | 52.6  | 37.3  |
|                                                                                  |             | MCWB   | 33.7  | 34.0  | 46.8  | 50.0  | 59.8  | 67.5  | 71.5  | 69.8  | 65.8  | 53.6  | 44.8  | 34.5  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 40.7  | 50.8  | 61.7  | 62.0  | 69.9  | 78.0  | 80.7  | 78.7  | 75.0  | 66.5  | 58.1  | 46.8  |
|                                                                                  |             | MCDB   | 47.3  | 57.9  | 70.7  | 76.3  | 81.8  | 89.1  | 88.5  | 87.5  | 84.4  | 75.7  | 67.9  | 51.9  |
|                                                                                  | 2%          | WB     | 38.1  | 40.9  | 55.7  | 58.1  | 67.1  | 74.9  | 78.0  | 76.1  | 72.6  | 61.4  | 52.5  | 40.0  |
|                                                                                  |             | MCDB   | 42.4  | 45.7  | 63.4  | 67.5  | 78.8  | 84.1  | 86.8  | 85.8  | 81.2  | 71.2  | 60.7  | 45.4  |
|                                                                                  | 5%          | WB     | 35.7  | 37.1  | 50.9  | 55.0  | 64.7  | 72.2  | 76.0  | 74.2  | 70.5  | 58.0  | 48.8  | 37.1  |
|                                                                                  |             | MCDB   | 38.6  | 41.2  | 58.7  | 64.2  | 75.6  | 81.8  | 85.3  | 82.8  | 78.8  | 67.4  | 57.5  | 40.5  |
|                                                                                  | 10%         | WB     | 33.5  | 34.1  | 46.1  | 51.5  | 62.4  | 69.9  | 73.8  | 72.0  | 68.0  | 54.9  | 44.9  | 34.6  |
|                                                                                  |             | MCDB   | 36.0  | 36.9  | 53.5  | 61.1  | 72.0  | 79.0  | 82.6  | 79.5  | 76.3  | 63.2  | 52.0  | 37.4  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 16.7  | 17.0  | 17.6  | 19.8  | 20.8  | 19.9  | 21.1  | 20.2  | 22.1  | 18.9  | 17.6  | 15.0  |
|                                                                                  |             | MCDBR  | 17.7  | 20.4  | 26.2  | 31.4  | 27.2  | 25.0  | 22.8  | 23.4  | 26.5  | 26.5  | 24.5  | 18.6  |
|                                                                                  |             | MCWBR  | 13.2  | 15.0  | 16.2  | 15.9  | 12.3  | 11.9  | 10.5  | 10.9  | 12.2  | 14.2  | 13.9  | 13.5  |
|                                                                                  | 5% WB       | MCDBR  | 16.7  | 19.6  | 24.8  | 25.1  | 22.4  | 21.5  | 20.6  | 20.8  | 21.8  | 23.0  | 23.0  | 16.8  |
|                                                                                  |             | MCWBR  | 13.1  | 14.4  | 15.9  | 15.4  | 11.8  | 11.5  | 10.9  | 11.1  | 12.3  | 13.6  | 13.6  | 12.6  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.265  | 0.281 | 0.315 | 0.360 | 0.377 | 0.389 | 0.397 | 0.402 | 0.359 | 0.322 | 0.286 | 0.269 |       |
|                                                                                  | taud        | 2.443  | 2.413 | 2.412 | 2.327 | 2.351 | 2.375 | 2.345 | 2.307 | 2.410 | 2.490 | 2.507 | 2.444 |       |
|                                                                                  | Ebn at Noon | 276    | 291   | 293   | 286   | 283   | 279   | 275   | 269   | 273   | 271   | 266   | 261   |       |
|                                                                                  | Edh at Noon | 22     | 28    | 32    | 38    | 38    | 38    | 38    | 38    | 32    | 26    | 21    | 20    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 580    | 915   | 1201  | 1513  | 1750  | 1949  | 2096  | 1734  | 1352  | 885   | 584   | 434   |       |
|                                                                                  | RadStd      | 64     | 102   | 121   | 143   | 129   | 131   | 76    | 146   | 101   | 113   | 56    | 41    |       |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (a)  | (b)                     | (c)   | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) | Station Only            |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (49 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page